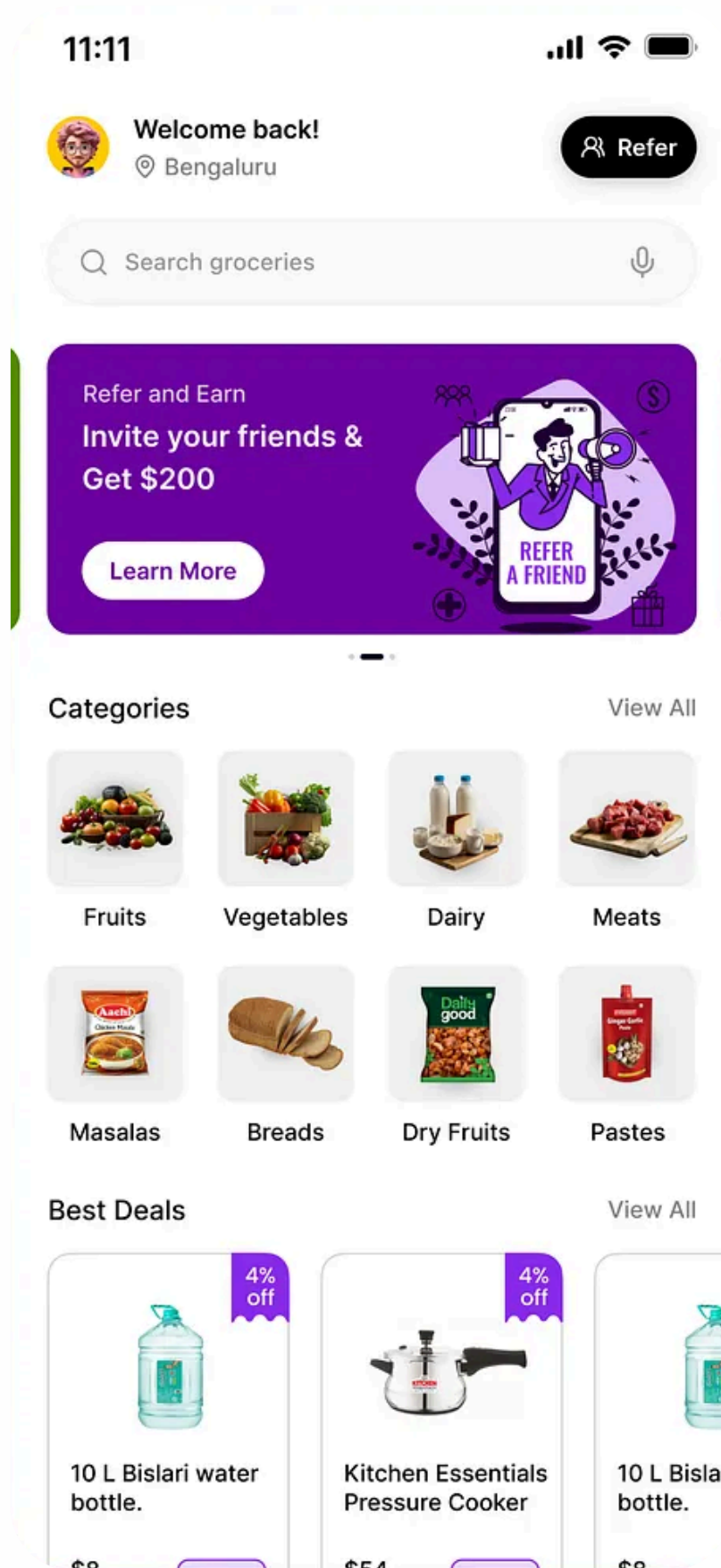




ZEPTO RETENTION

From User Research to a Data-Backed Replenishment Product

18 user interviews
80k orders analyzed
SQL + Excel validation
Product strategy & MVP

Independent Product Case Study

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Business Analytics · Product · Strategy

EXECUTIVE SUMMARY

From diagnosing churn to designing a specific, testable product intervention.



BUSINESS QUESTION

Why do users churn, where does the journey break, and which product lever could improve repeat purchasing?

APPROACH

18 user interviews · 80K orders · SQL analysis · 5 hypotheses

KEY FINDING

No single behavioural driver explained churn. Membership showed the strongest retention association, while several intuitive hypotheses were not supported or could not be directly validated.

PRODUCT DECISION

Rather than building a generic churn-reduction feature, I focused on repeat-purchase friction among Routine Rebuyers.

SMART REPLENISHMENT

A proposed assistant that identifies repeated, predictable purchase patterns, surfaces relevant items when they are likely due, and lets users review and reorder.

VALIDATION PLAN

Randomized A/B test → Normal Zepto vs. Smart Replenishment
Primary KPI: Repeat Purchase Rate



Why do users churn, where does the customer journey break, and which product levers could improve retention and repeat purchasing?

JOURNEY: Acquire → First Order → Repeat Purchase → ??? → Churn

Retention gap to investigate

How I approached it

01 — Listen

18 user interviews

02—Hypothesize

5 retention hypotheses

03 — Validate

80K+ orders + SQL analysis

04 — Decide

Data-backed product opportunity



WHAT USERS DESCRIBED

The interviews surfaced four recurring themes:

- **Trust & reliability** — Wrong or damaged orders, refund issues and poor issue resolution could affect future usage.
- **Inventory availability** — Users were frustrated when preferred products or brands became unavailable.
- **Price & value** — Coupons, delivery charges and competitor prices influenced where users ordered.
- **Repeat purchasing** — Users regularly bought recurring essentials, but rebuilding the same order could be inconvenient.

These were useful signals, but I didn't want to assume that a problem mentioned in an interview was necessarily causing churn.

HOW I GOT TO THE HYPOTHESES

I didn't want to assume that these were necessarily causing churn. I converted the recurring themes into hypotheses and tested them against the order data.

Hypothesis	What I tested
H1 — Trust & Reliability	Does delivery behaviour differ between retained and churned users?
H2 — Inventory	Does availability-related behaviour explain churn?
H3 — Price & Value	Does coupon/price behaviour relate to retention?
H4 — Habit Formation	Do recurring purchase patterns differentiate churn?
H5 — Membership	Does membership status relate to retention?

WHAT USERS ACTUALLY TOLD ME

I spoke to 18 users about how they order, what makes them return, and what makes them consider another platform.



Theme	What I heard	Example from interviews	Why I investigated it
Trust & Reliability	Wrong/damaged items, refunds and service issues came up as reasons for frustration.	Priya paused after a wrong medicine; Ritu after damaged formula.	Could poor service experiences reduce repeat usage?
Inventory	Users expected frequently purchased products to be available.	Priya found items shown as available were not actually available.	Could availability problems interrupt repeat purchasing?
Price & Value	Users compared coupons, delivery fees and offers across platforms.	Aman chose the lowest delivery fee; Imran followed coupons.	Could price-sensitive behaviour be linked to churn?
Repeat Purchasing	Some users repeatedly bought the same essentials and wanted less effort in reordering.	Users mentioned recurring essentials and rebuilding familiar orders.	Could repeat-purchase friction be a product opportunity?

The interviews gave me several plausible explanations for weak retention
But they didn't tell me which one was actually driving churn.
That became the reason for the quantitative validation.



HOW I TURNED INTERVIEW THEMES INTO HYPOTHESES



After affinity mapping the interview observations, I grouped similar pain points into recurring themes. I then turned those themes into hypotheses that could be checked against the order data.

Hypothesis	Why I considered it	What I wanted to find out
H1 — Trust & Reliability	Users described wrong/damaged orders and service issues.	Does delivery behaviour differ between retained and churned users?
H2 — Inventory & Stock	Availability problems came up repeatedly.	Does availability-related behaviour explain churn?
H3 — Price & Value	Users compared coupons, fees and offers.	Does coupon/price behaviour relate to retention?
H4 — Habit Formation	Some users described repeatedly buying the same everyday essentials.	Do recurring purchase patterns differentiate churn?
H5 — Membership & Loyalty	Membership was added as a separate behavioural segment to test as a potential retention differentiator.	Does membership status relate to retention?



The interviews told me what might matter. The next step was to see whether those patterns actually differentiated churn in the data.

HOW I TESTED THE HYPOTHESES

I used the data to check whether the behaviours suggested are actually different between retained and churned users.



WHAT I ANALYSED

80K+
order records

Unit of analysis

User-level behaviour derived from order history

Primary comparison

Retained vs. churned users

Key measures

- Order frequency
- delivery behaviour
- coupon usage
- recurring purchases
- membership

HOW I COMPARED BEHAVIOUR

FROM HYPOTHESIS → TEST

Hypothesis	Behaviour I compared
Trust & Reliability	Delivery behaviour / orders per user
Inventory	Availability signals
Price & Value	Coupon usage / order behaviour
Habit Formation	Recurring-category behaviour
Membership	Churn rate / orders / revenue

For each hypothesis, I compared the relevant measure between retained and churned users.

TOOLS

SQL

Used to:

- aggregate order-level data
- calculate user-level metrics
- compare segments
- test hypotheses

EXCEL

Used to:

- organize outputs
- compare results
- consolidate evidence
- build the Master Evidence Table

For each hypothesis, I looked for a meaningful difference between retained and churned users.

WHAT THE DATA SHOWED

Not every interview finding translated into a meaningful difference in churn.



5 HYPOTHESIS RESULTS:

Hypothesis	What the data showed	Verdict
H1 – Trust & Reliability	Delivery behaviour did not meaningfully differ between retained and churned users.	Not supported
H2 – Inventory	Direct stock-out data wasn't available; search/cart behaviour was used only as an indirect signal and didn't distinguish churn.	Not validated
H3 – Price & Value	Coupon users were more active, but coupon dependency didn't meaningfully explain churn.	Partially supported
H4 – Habit Formation	Recurring purchasing was real, but churn was almost identical: 32.85% vs 31.91% .	Not a churn differentiator
H5 – Membership	Premium churn was 15.34% vs 40.41% for free users.	Strongest observed signal

ONE IMP CHECK

Premium and free users had almost identical order frequency:

7.99 vs 8.01
orders/user
and revenue/user
₹5,196 vs ₹5,189

THE CLEAREST DIFFERENCE:

15.34% vs. **40.41%**
Premium churn Free-user churn

The analysis didn't give me one universal reason for churn. Membership showed the clearest retention association, while several intuitive explanations were either weak, unsupported, or impossible to directly test with the available data.



WHAT I TOOK FROM THE ANALYSIS

Because the analysis didn't point to a single root cause for churn, I focused on a real, addressable product problem.

1 — Churn doesn't have one proven explanation

Several interview themes were plausible, but the quantitative analysis did not show one behaviour that explained churn across users.

H1 not supported · H2 not validated · H3 partial · H4 not a churn differentiator

2 — Membership showed the strongest retention association

Premium users had substantially lower observed churn than free users (15.34% vs. 40.41%), although the analysis does not establish that membership causes the difference.

Next question: What is driving that association?

3 — Repeat-purchase friction was still a product opportunity

Recurring purchasing was real, even though it didn't meaningfully differentiate churn. Users already knew what they wanted to buy again; the opportunity was to reduce the effort of finding and rebuilding those purchases.

THE PRODUCT DECISION

I chose not to build a generic “churn reduction” feature. I focused on making repeat purchases easier for Routine Rebuyers. Membership remains a separate retention opportunity to investigate further.

THE PRODUCT PROBLEM



For users who repeatedly buy the same essentials, reordering still requires rebuilding the purchase each time.



THE USER WHO ARE WE SOLVING FOR?

ROUTINE REBUYER

Repeated

Buys the same / similar products again

Predictable

Purchase behaviour follows a recognizable pattern

Intent is already there

Usually knows what they want to buy again

Not every Zepto user, only users with a predictable repeat-purchase pattern.

THE FRICTION THE CURRENT EXPERIENCE

- Remember
- Search
- Find
- Add
- Rebuild
- Checkout

The purchase is familiar.
The process isn't.

THE PRODUCT OPPORTUNITY WHAT CAN WE CHANGE?

Recognise

Identify a predictable repeat pattern



Review

Show the user what is ready to replenish



Reorder

Let them confirm and reorder with minimal effort

The product reduces effort.
It doesn't decide what the user should buy.

PRODUCT PROBLEM

How might Zepto recognise predictable repeat-purchase behaviour and make the next replenishment easier, without adding friction or taking control away from the user?



WHY SMART REPLENISHMENT?

The analysis narrowed the opportunity: make repeat purchases easier for users with predictable buying patterns.

01 — WHAT THE ANALYSIS RULED OUT

- Trust & reliability → Not supported
- Inventory → Not validated
- Habit formation → Not a churn differentiator
- Coupon behaviour → Only partially supported

02 — WHAT WAS STILL ACTIONABLE

Recurring purchasing was real. Users repeatedly bought familiar products/categories, creating an opportunity to reduce the effort involved in repeating those purchases.

03 — THE PRODUCT CHOICE

Focus on Routine Rebuyers. Users already know what they want.
The opportunity is to remove the repetitive work between intent and checkout.

Membership showed the strongest retention association, but the analysis does not establish causality.
It remains a separate retention question.

SMART REPLENISHMENT ASSISTANT MODEL

Making repeat purchases easier for users with predictable buying patterns.



1 — RECOGNISE

IDENTIFY REPEAT-PURCHASE PATTERNS

The system identifies products that a user repeatedly purchases with a sufficiently predictable pattern. *Illustrative examples:*

-  Coke + Chips
purchased repeatedly
-  Milk + Household essentials
recurring pattern

Repeated + predictable → eligible

2 — REVIEW

YOUR USUALS

-  Coke ✓
-  Chips ✓
-  Milk ✓

3 items ready to replenish

Review basket →

- Edit
- Remove
- Change quantity

3 — REORDER

ONE TAP REORDER

-  Coke ✓
-  Chips ✓
-  Milk ✓

3 items · ₹XXX

REVIEW & REORDER →

User confirms the basket and places the order.

Next purchase → updates the pattern

**THE PRODUCT DOESN'T DECIDE FOR THE USER.
IT PREPARES THE REORDER.**

PURCHASE HISTORY → PATTERN DETECTED → YOUR USUALS → REVIEW → REORDER → LEARN

HOW IT WORKS

A decision model identifies repeat patterns, decides when a reorder is relevant, and learns from what the user actually does.



THE CORE DECISION MODEL

Repeated + Predictable + Due → Replenish

Not every past purchase becomes a recommendation.

01 — DETECT THE PATTERN

The system looks at purchase history to identify:

- repeated purchases
- consistency of purchase behaviour
- approximate replenishment interval

Repeated + sufficiently predictable → candidate

02 — DECIDE IF IT'S DUE

The system compares the user's expected purchase window with the latest purchase.

Example: Usual interval: **~8 days**
Last purchase: **14 days ago**
→ **Item enters the replenishment window**

But: Bought only once → **Don't recommend**
Highly irregular → **Wait for more evidence**
Bought recently → **Don't surface**

03 — PREPARE, DON'T AUTO-ORDER

Eligible items are surfaced in **Your Usuals** and placed into a reviewable basket.

User can:

- remove
- change quantity
- add something else
- ignore

No automatic ordering.

That user-control decision is explicitly part of your product logic.

04 — LEARN FROM ACTUAL BEHAVIOUR

The system doesn't permanently assume the original pattern.

Expected: 8 days
Actual: 5 days
→ update the pattern

Expected: 8 days
Actual: 14 days
→ shift the window

No purchase:
→ don't immediately assume the user stopped wanting it

Current project proposes the decision logic, not a production SKU-level recommender.
The available dataset is category-level; item-level data would be required for SKU-level recommendations.

USER JOURNEY

The experience stays familiar; Smart Replenishment removes the repetitive work.



01 — NORMAL PURCHASE

01

User buys as usual

No change to the existing experience.

04 — REVIEW

04

User checks the basket

Edit quantity · Remove or Add something else

02 — PATTERN BUILDS

02

Purchase history builds a repeat pattern

No setup required from the user.

05 — REORDER

05

User confirms the purchase

Faster than rebuilding the basket from scratch.

03 — YOUR USUALS

03

Relevant purchases appear when likely due

The user sees familiar items ready to replenish.

06 — LEARN

06

Actual behaviour updates the pattern

Earlier, later or skipped purchases inform future recommendations.

NO MANUAL SETUP. NO AUTO-ORDERING. THE USER STAYS IN CONTROL.

PRODUCT FLOW

How the product moves from a user's purchase history to a replenishment decision.



1. Purchase History

Past orders provide the behavioural signal.



2. Identify Recurring Patterns

Find purchases that repeat over time.



3. Determine Replenishment Eligibility

Check whether the pattern is sufficiently repeated and predictable.



4. Build “Your Usuals”

Create the set of purchases that qualify.



5. Surface Replenishment

Show the eligible usuals when the user is likely due.



6. Review & Reorder

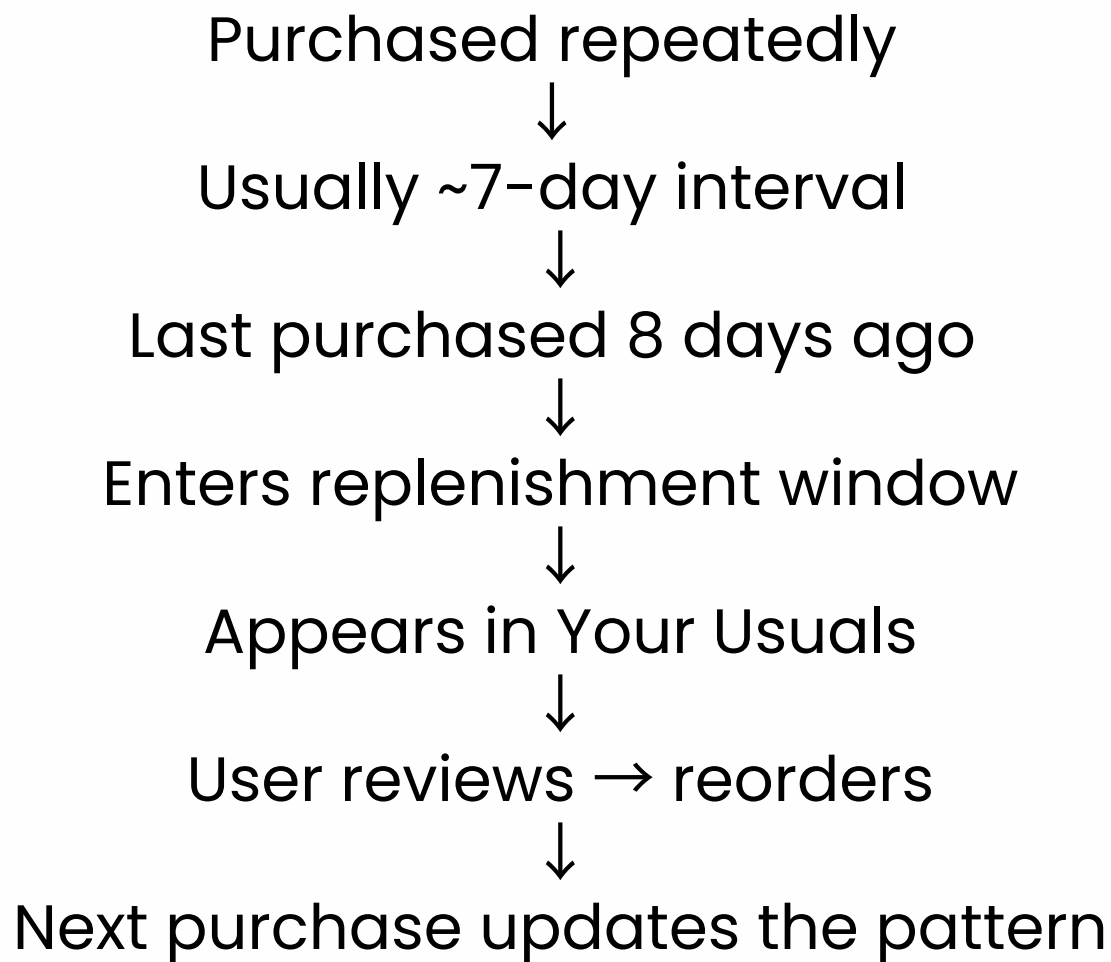
User edits the basket and confirms the order.



7. Update Pattern

Actual behaviour becomes the next data point.

Illustrative Example: Milk

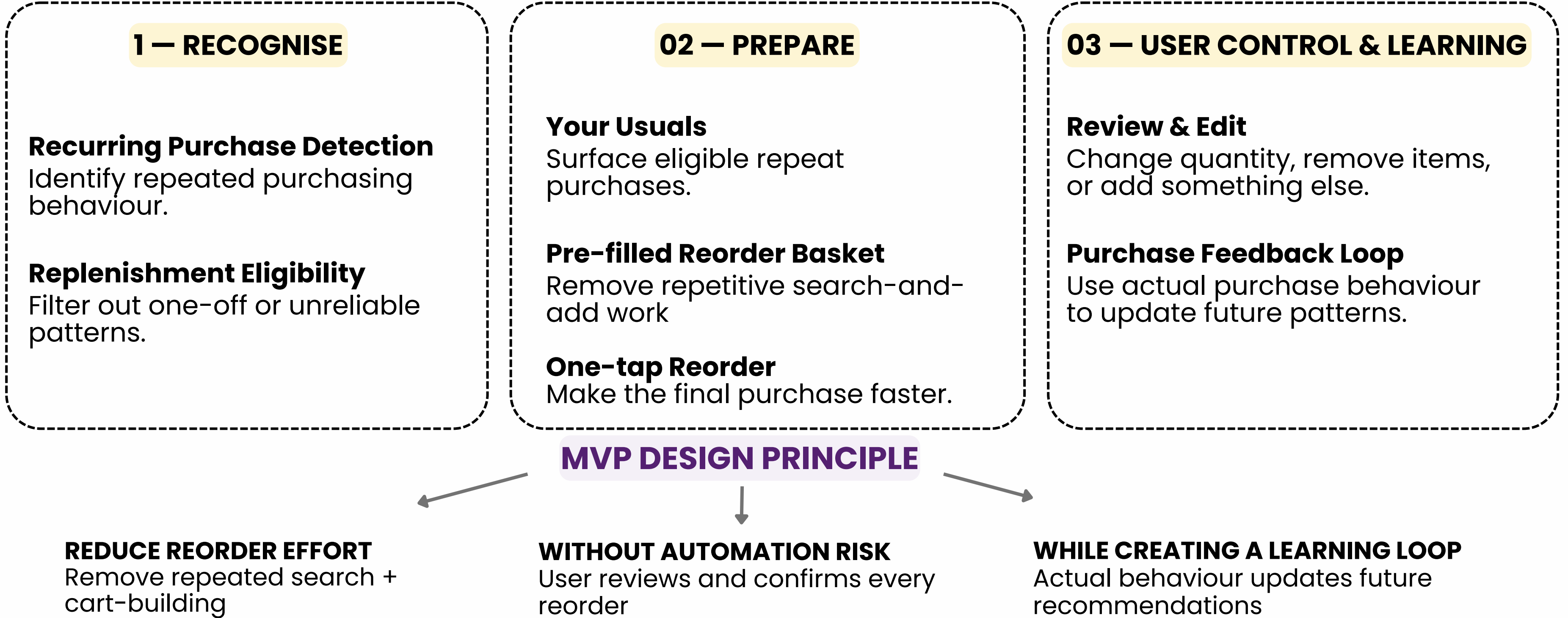


Illustrative example of the proposed decision logic

MVP FEATURES



The MVP focuses on the smallest set of features needed to reduce reorder friction and test the core behaviour.



Deliberately excluded from MVP:
Reminders · Scheduled purchases · Subscriptions · Advanced personalization



PRODUCT LOGIC & EDGE CASES

The recommendation should adapt to behaviour, not blindly repeat past purchases.

WHEN SHOULD AN ITEM BE RECOMMENDED?

RECOMMEND

Repeated + predictable + likely due

DON'T RECOMMEND YET

Bought once

Highly irregular pattern

DON'T SURFACE

Purchased recently

Product unavailable

WHAT IF BEHAVIOUR CHANGES?

EXPECTED → ACTUAL → UPDATE

Bought earlier

Expected: 8 days → Actual: 5 days
→ Update the pattern

Bought later

Expected: 8 days → Actual: 14 days
→ Shift the replenishment window

Skipped purchas

No purchase

→ Don't immediately assume the user stopped wanting it

USER CONTROL IS A HARD RULE

Remove · Change quantity · Add something else · Ignore

No automatic ordering.

Current dataset is category-level; SKU-level recommendations would require item-level purchase and quantity data.



PRIORITIZATION

I prioritized the core reorder problem before adding automation or deeper personalization.

P0 — CORE MVP

Solve the problem first

- Recurring purchase detection
- Replenishment eligibility
- Your Usuals
- Pre-filled reorder basket
- Review & edit
- One-tap reorder
- Purchase feedback loop

Why: Directly addresses reorder friction and is needed to test the core hypothesis

P1 — NEXT ITERATION

Make replenishment proactive

- Smart replenishment reminders
- Personalized timing
- Better basket grouping / personalization
- User reminder preferences

Why: Adds proactive value only after the basic reorder experience proves useful.

P2 — FUTURE

Add automation & deeper personalization

- Scheduled replenishment
- Flexible subscriptions
- Skip / pause / reschedule
- Advanced personalization
- Membership integration

Why: Higher complexity and should come after the core behaviour is validated.

I deliberately started with assisted replenishment rather than subscriptions or fully automated ordering.



The product hypothesis would be tested through a randomized A/B experiment.

PRODUCT HYPOTHESIS

If eligible Routine Rebuyers get an easier way to reorder their usual purchases, they will be more likely to make another purchase than similar users using the normal Zepto experience.

A/B TEST:

CONTROL	VS.	TREATMENT
Normal Zepto		Normal Zepto + Smart Replenishment

PRIMARY KPI

Repeat Purchase Rate

SECONDARY

- Replenishment → reorder conversion
- Time to reorder
- Orders / user
- Replenishment basket adoption

GUARDRAILS

- Recommendation dismissal
- Item removal / opt-out
- Cancellation / refund behaviour

Eligible Routine Rebuyers → defined observation window → compare repeat behaviour
Historical analysis is observational; the A/B test is the proposed causal validation.

PRODUCT ROADMAP

Start with assisted replenishment; increase automation as the core behaviour is validated.



Phase 1 — MVP

Reduce reorder friction

Recurring detection
Eligibility
Your Usuals
Review & reorder
Feedback loop

Outcome:

Validate whether easier reordering increases repeat purchasing.

PHASE 2 — PROACTIVE REPLENISHMENT

Help users at the right time

Smart reminders
Personalized timing
Better replenishment ranking
Reminder preferences

Outcome:

Move from users discovering their usuals to Zepto proactively helping them.

PHASE 3 — SCHEDULED PURCHASES

Support highly predictable needs

Scheduled replenishment
Flexible subscriptions
Skip / pause
Quantity adjustment
Rescheduling

Outcome:

Reduce effort further for predictable recurring needs.

PHASE 4 — PERSONALIZATION + MEMBERSHIP

Build a broader retention experience

Advanced prediction
Personalized replenishment
Changing-pattern handling
Membership integration

Outcome:

Expand replenishment into a more personalized retention ecosystem.

ASSIST → PROACT → AUTOMATE → PERSONALIZE

I deliberately started with assisted replenishment rather than subscriptions or fully automated ordering.



FINAL RECOMMENDATION

A focused retention opportunity, validated in stages rather than assumed.

01 — WHAT THE ANALYSIS TOLD ME

There was no single proven behavioural driver of churn. Membership showed the strongest observed association with retention, while several other hypotheses were weak, unsupported, or could not be directly validated.

02 — WHAT I CHOSE TO BUILD

SMART REPLENISHMENT

Focus on Routine Rebuyers who already show repeated, predictable purchasing behaviour. The product reduces the effort of rebuilding familiar purchases while keeping the user in control.

03 — HOW I WOULD VALIDATE IT

A/B test. Normal Zepto vs. Normal Zepto + Smart Replenishment

Primary KPI: Repeat Purchase Rate

Guardrails: recommendation dismissal, opt-out/removal, cancellation/refund behaviour.

ONE BIG CONCLUSION: Don't try to "solve churn" with one feature. Identify a specific user behaviour, remove a real friction, and prove that the intervention changes behaviour.



A1 — RESEARCH EVIDENCE

18 interviews were used to generate hypotheses, not to prove causality.

Theme	Interview signal	Users mentioning it	What it led me to test
Price & value	Coupons, fees, offers	10/18	H3
Speed & ETA	Convenience / delivery speed	8/18	Context, not directly tested
Inventory	Availability problems	7/18	H2
Trust / reliability	Wrong/damaged orders, refunds	11/18	H1
Support	Issue resolution	6/18	Research context
Repeat purchasing	Recurring essentials / reorder friction	5/18	Product opportunity



A2 — INTERVIEW AFFINITY MAP

RAW OBSERVATIONS

Wrong medicine
Damaged formula
Refund took days
Lowest delivery fee
Items unavailable
Buys same essentials
Forgets milk
Compares offers



AFFINITY GROUPS

Trust & Reliability
Price & Value
Inventory
Repeat Purchasing



QUESTIONS TO VALIDATE

Could this differentiate churn?



A3 — DATASET & ANALYTICAL SETUP

ORDERS	USERS	SESSIONS	SUBSCRIPTIONS
80,000 rows	10,000 users	50,000 sessions	10,000 users
Order behaviour	Churn / profile attributes	Search → cart → checkout behaviour	Membership status

UNIT OF ANALYSIS

User-level behaviour

Order/session records were aggregated into user-level metrics before comparing retained and churned users.

CORE OUTCOME

Churned vs. retained

Why user-level?

A user can place many orders, so analysing raw order rows would overweight frequent users. The hypothesis tests therefore use user-level aggregates where appropriate.

A4 — CHURN & USER-LEVEL METRICS

10,000 users

→ 9,997 users with orders

→ 9,969 users with ≥ 2 orders

Churned: $3,281 / 10,000 = \mathbf{32.81\%}$

Retained: $6,719 / 10,000 = \mathbf{67.19\%}$

Example user-level measures

- Orders per user
- Revenue per user
- Coupon usage
- Recurring-category behaviour
- Delivery behaviour
- Membership status



APPENDIX A5 – H1 & H2: WHAT THE DATA COULD / COULD NOT TEST

H1 – TRUST & RELIABILITY

Question:
Does delivery behaviour differ between retained and churned users?

Available signal:
Delivery time / order behaviour.

Finding:
No meaningful difference.

Verdict:
Not supported

Limitation
The dataset does not contain:

- wrong-item incidents
- damaged items
- refund delays
- support failures

H2 – INVENTORY

Question:
Does availability-related behaviour explain churn?

Available evidence:
No direct stock-out field. Used session behaviour as an indirect signal.

Finding:
Search/cart behaviour did not meaningfully differentiate churn.

Verdict:
Not validated



A6 — H3 + H4 + H5

H3 — PRICE & VALUE

Coupon behaviour	Churn
No coupon	~33.83%
Low	~33.23%
High	~32.50%
Very high	~27.60%

Coupon users were generally more active, but coupon dependency did not provide a strong explanation for churn.
Verdict: Partially supported

H4 — HABIT FORMATION

Recurring-category behaviour
Repeat category: 32.85% churn
No repeat: 31.91% churn
Difference is small.
Verdict: Not a churn differentiator

H5 — MEMBERSHIP

Premium: 15.34% churn
Free: 40.41% churn
Orders/user: 7.99 vs 8.01
Revenue/user: ₹5,196 vs ₹5,189
Membership is strongly associated with lower observed churn, but the analysis does not establish causality.
Verdict: Strongest observed signal



A7 — MASTER EVIDENCE TABLE

Hypothesis	Evidence	Verdict	What it means
H1 Trust	Delivery behaviour showed no meaningful difference	Not supported	Interview concern not validated by available proxy
H2 Inventory	No direct stock-out data; indirect session signal weak	Not validated	Need inventory-level data
H3 Price	Coupon users more active; dependency weakly related to churn	Partial	Activity ≠ retention driver
H4 Habit	32.85% vs 31.91% churn	Not differentiator	Repeat behaviour exists, but doesn't explain churn
H5 Membership	15.34% vs 40.41% churn	Strongest signal	Strong association; causality unknown

Now, I stopped looking for one universal churn cause and moved toward a specific, addressable repeat-purchase problem.

APPENDIX A8 — REPLENISHMENT ELIGIBILITY LOGIC

01 — REPEAT BEHAVIOUR

Has the user purchased the same product/category multiple times?

02 — PURCHASE INTERVAL

Example:

8 → 7 → 9 days

03 — CONSISTENCY

8, 8, 9, 7 → predictable vs. 2, 31, 5, 47 → irregular

04 — RECENCY / DUE WINDOW

Typical interval ≈ 8 days

Last purchase approaching that window
→ candidate

Repeated? → Predictable? → Due? → Eligible

Exact thresholds are intentionally not hard-coded in this case study; they would be calibrated using historical data and experimentation before production.

A9 — EXPERIMENT DECISION RULES

WHAT HAPPENS AFTER THE TEST?

POSITIVE

Higher Repeat Purchase Rate

- acceptable guardrails

→ Continue / expand testing

NO MEANINGFUL IMPROVEMENT

Users engage but repeat behaviour doesn't improve

→ Investigate:

- recommendation quality
- timing
- basket composition
- user value

NEGATIVE

High dismissal / removal / cancellations / refunds

→ Don't scale

Rework eligibility logic or experience.

A10 — RESEARCH METHODOLOGY

18 users interviewed

Who

- Existing quick-commerce users
- Different usage frequencies/needs

What I explored

- Ordering habits
- Reasons for repeat purchase
- Switching behaviour
- Pain points
- Coupons / availability / reliability
- Reordering behaviour

How

- Semi-structured interviews
- Open-ended questions
- Notes captured → affinity mapping → themes

While Interviews generated hypotheses,
I used SQL analysis to test them.

A11 — ROOT CAUSE ANALYSIS

OBSERVED RETENTION PROBLEM



Potential causes

Trust / Reliability

Wrong or damaged orders → reduced confidence

Inventory

Unavailable expected products → purchase interruption

Price / Value

Coupons, fees, competitor comparison → switching

Habit / Repeat Purchase

Repeated essentials → opportunity for easier replenishment

Membership

Different engagement/retention behaviour → separate hypothesis



Testable hypotheses H1–H5

A12 — PRODUCT REQUIREMENTS

Problem

Routine Rebuyers repeatedly rebuild familiar purchases.

Target user

Users with repeated + predictable purchasing behaviour.

Core user need

Replenish familiar purchases with less effort while retaining control.

MVP requirements

1. Detect recurring behaviour
2. Determine replenishment eligibility
3. Surface Your Usuals
4. Pre-fill reorder basket
5. Allow review/edit
6. Enable reorder
7. Learn from subsequent behaviour

Out of scope

- Fully automated ordering
- Subscriptions
- Advanced personalization



A13 — RISKS & TRADE-OFFS

Risk	Why it matters	Mitigation
False recommendation	User loses trust	Require repeated + predictable behaviour
Recommendation too early	User just purchased	Apply recency / due-window check
Irregular purchases	Poor prediction	Don't recommend until pattern strengthens
User feels controlled	Product becomes intrusive	Review/edit + no auto-ordering
Behaviour changes	Old pattern becomes wrong	Feedback loop updates pattern
Product unavailable	Bad reorder experience	Don't surface unavailable items
Data limitation	Category ≠ SKU	Production needs item-level data

APPENDIX — SQL ANALYSIS



Each hypothesis was translated into a specific SQL analysis before being included in the final evidence table.

Analysis	What the SQL tested	Output used in project
H1 — Trust & Reliability	Delivery behaviour across retained vs churned users	No meaningful difference
H2 — Inventory	Search → cart behaviour + churn comparison	No meaningful differentiation; no direct stock-out field
H3 — Price & Value	User-level coupon behaviour vs churn	Coupon dependency did not explain churn
H4 — Habit Formation	Repeated-category purchasing vs churn	32.85% vs 31.91%
H5 — Membership	Premium vs Free churn, orders/user, revenue/user	15.34% vs 40.41% churn
Sessions	Funnel/search/cart behaviour by retention status	Behaviour broadly similar

SQL WORKFLOW

- Joins / aggregations
- User-level metrics
- Retained vs churned comparison
- Hypothesis verdict

[Full SQL files for H1–H5 and session analysis are included in the project GitHub repository.](#)